LESSON 07 · MISSION GUIDE

The Toolbox

Modules, the standard library, virtual environments and pip: using code you didn't write.

From **Mo, IT**
To **Jen**
Subject **You keep copying the same functions**

Jen. I had a look at your `my_work` folders. Lovely stuff. Also: `clean_money()` appears in eleven files, three of them slightly different. When Priya changes her mind about pence, you'll fix it eleven times and miss one. I've seen this film. Dave was in it.

Today: put your functions in a module and import them. Meet the standard library, written by people with more patience than either of us. Then `pip`, which installs packages other people wrote, inside a virtual environment, so nothing you install can break your laptop or my weekend. Next week it installs NumPy and pandas, and MACRO-9000 gets nervous. — Mo

What you'll be able to do after today

- [ ] Create and activate a virtual environment, and explain to Gerald what it's for in one sentence
- [ ] Install packages from a `requirements.txt` with `pip`, and check what's installed
- [ ] Use `import` three ways: `import math`, `from datetime import date`, `import paytools`
- [ ] Borrow from the standard library: `math`, `statistics`, `datetime`, `random`
- [ ] Put your own functions in a module and import them from another file
- [ ] Use `if __name__ == "__main__":` so a module is quiet when imported, chatty when run

Four big ideas

1. A module is a file	`paytools.py` is the module `paytools`. `import paytools` runs the file once and gives you its functions as `paytools.monthly()`.
2. The standard library is already installed	`math`, `statistics`, `datetime`, `random`, `csv`, `pathlib` … no pip needed. `help(module)` in the REPL is the manual.
3. pip installs everything else	Hundreds of thousands of packages on PyPI. `pip install rich` downloads, installs, and makes `import rich` work.
4. A venv is a toolbox per project	Packages installed with the venv active go into `.venv/`, not into Windows. Delete the folder, and it's as if nothing happened.

Three things that bite

- Forgetting to activate. "No module named rich" usually means the venv is asleep. Look for `(.venv)` in the prompt.
- A file called `random.py` (or `csv.py`, `math.py`) in `my_work/` hides the real module.
- Committing `.venv/`: thousands of files that belong to your laptop. `.gitignore` hides it; check `git status`.

Why this matters at work

- **Don't write what exists.** Dates, statistics, Excel files, charts: somebody has done it, tested it and documented it. Plug it in.
- **One function, one place.** A module is the difference between "fix it once" and "find all eleven copies".
- **Reproducible setup.** `requirements.txt` + a venv: the same toolbox on any laptop in two commands. No "works on Dave's machine".

How today works

15 MIN
Lecture
Lars makes a venv, installs a package, writes a module, live

60 MIN
Exercise sheet
Seven tickets, from your own toolbox to a package you've never seen

15 MIN
Dave's Leftovers #7
His team planner, from the week he discovered import

The inbox

#	FROM	TICKET	YOU'LL USE
1	Mo	A toolbox of your own	`python -m venv`, `activate`, `pip install -r`
2	Gerald	Borrow from the standard library	`math.ceil`, `statistics.mean/median/stdev`
3	Priya	Dates without tears	`date`, `timedelta`, `strftime` codes
4	Mo	Your own module	`paytools.py`, `import paytools`, docstrings
5	Siobhan	Biscuit's raffle	`random.seed/choice/sample/shuffle`
6	Gerald	Something you didn't write	`rich`, reading docs
7 ⭐	Mo	The main guard	`if __name__ == "__main__":`

You're done when...

- [ ] Your prompt shows `(.venv)` and `pip list` shows `rich`
- [ ] `my_work/` has `paytools.py` and `ex2` to `ex6` (and `ex7`), each printing the expected output
- [ ] Dave's planner prints the four expected lines, one commit per bug
- [ ] Everything is pushed to `lesson-07`, `.venv/` is not, and you've opened a pull request



Next lesson: Whole Columns at Once. Your first `pip install numpy`. Instead of looping over 200 salaries, you'll do the maths on the whole column in one line, the way you drag a formula down in Excel, except it's instant and never off by a row.